

40.520 Stochastic Models

Martingale Theory

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Outline

1. Review on Conditional Expectation
2. Martingale Definition
3. Martingale Convergence Theorem
4. Martingale Stopping Theorem
5. Martingale Inequalities

Conditional Expectation

Conditional Expectation

- Given two events A and B with $P(A) > 0$, define the conditional probability $P(B|A) = \frac{P(A \cap B)}{P(A)}$.
 - If $P(A) = 0$, the usual convention is that $P(B|A) = 0$.
- Given a discrete random variable Y , we define its conditional expectation with respect to the event A by

$$E[Y|A] = \sum_y y \cdot P(Y = y|A)$$

Conditional Expectation

- When the event $A = \{X = x\}$ where X is another discrete random variable, we define the function

$$f(x) = E[Y|X = x]$$

- We define $E[Y|X] = f(X)$, which means that $E[Y|X]$ is itself a random variable.

Properties

- If $X \perp Y$, then $E[Y|X] = E[Y]$
- If $Y = g(X)$, then $E[YZ|X] = YE[Z|X]$
- Tower Property:

$$E[E[X|Y, Z]|Y] = E[E[X|Y]|Y, Z] = E[X|Y]$$

and in particular, $E[E[Y|X]] = E[Y]$

- $E[C|Y] = C$ for any constant number C

Definition of Martingale

What is a Martingale?

A stochastic process $\{Z_n, n = 1, 2, \dots\}$ is said to be a martingale process with respect to $\{X_1, X_2, \dots\}$ if

$$E[Z_n | X_1, \dots, X_{n-1}] = Z_{n-1} \text{ for all } n.$$

A martingale can be viewed as a generalized version of a fair game. We have

$$E[Z_n] = E[Z_{n-1}].$$

Key insight: The best forecast for tomorrow is what we know today — no drift!

Example: A Fair Play

- A gambler plays a game repeatedly. At each play he wins \$1 with prob. 50% and loses \$1 with prob. 50%. The plays are independent of each other.
- Let X_n be the outcome of each play and

$$Z_n = \sum_{i=1}^n X_i$$

Show $Z_n, n \geq 1$ is a martingale.

Example: Product of Random Numbers

- Let $\{X_1, X_2, \dots\}$ be a sequence of independent random variables with

$$E[X_i] = 1$$

- Show

$$Z_n = \prod_{i=1}^n X_i, \quad n \geq 1,$$

is a martingale.

Example: Variance of a Sum as a Martingale

- Let $\{X_1, X_2, \dots\}$ be a sequence of independent and identically distributed random variables with

$$E[X_i] = 0 \quad \text{and} \quad \text{Var}[X_i] = \sigma^2$$

- Let

$$Y_n = \left(\sum_{i=1}^n X_i \right)^2 - n\sigma^2$$

- Show $\{Y_n, n = 1, 2, 3, \dots\}$ is a martingale.

Example: Pólya's Urn

Start with r red balls, b blue balls. At each step:

- Draw a ball uniformly
- Return it + add another ball of the same color

Let R_n = number of red balls after n steps. Let $Z_n = \frac{R_n}{n + r + b}$ = proportion of red balls.

Result: The ratio does not explode toward the color with more initial balls.

Martingale Convergence Theorem

Martingale Convergence

- Does a martingale converge?
- Will Z_n approach a limit as $n \rightarrow \infty$?

Not all martingales converge — we need an extra condition.

Martingale Convergence Theorem

Theorem (Martingale Convergence Theorem)

Let $(Z_n)_{n \geq 1}$ be a martingale. If

$$\sup_n \mathbb{E}[|Z_n|] < \infty$$

then:

1. There exists a random variable Z_∞ with $\mathbb{E}[|Z_\infty|] < \infty$
2. $Z_n \rightarrow Z_\infty$ **almost surely**

Key Insight: If the expected absolute value doesn't blow up, the martingale must settle down.

Example: Symmetric 1D Random Walk

Let $Z_n = \sum_{i=1}^n X_i$ with $P(X_i = \pm 1) = 1/2$

- Is Z_n a martingale?
- Does it converge?

Example Gambler's Ruin

A gambler starts with \$ k , aims for \$ N , or goes broke. Let Z_n = wealth at time n .

- Is Z_n a martingale?
- Does it converge? What are the possible limits?

Martingale Stopping Theorem

Stopping Times

- The positive integer-valued (possibly infinite) random variable N is said to be a stopping time for a process $\{Z_n, n = 1, 2, \dots\}$ if the occurrence of event $\{N \leq n\}$ is determined by the value of $\{Z_1, Z_2, \dots, Z_n\}$.

Example: Hitting Time of a Process

Consider a process $\{Z_n, n = 1, 2, \dots\}$. Let

$$N = \min\{n : Z_n \geq 1\},$$

Is N a stopping time?

Example: Maximum Time

Consider a process $\{Z_n, n = 1, 2, \dots\}$. Let

$$N = \min \left\{ n : Z_n = \max_{1 \leq i \leq 10} Z_i \right\}$$

i.e., N is the first time that the process reaches the largest value of $Z_1, Z_2, \dots, Z_{10}$.

Is N a stopping time?

Martingale Stopping Theorem

Theorem (Martingale Stopping Theorem)

If $\{Z_n, n = 1, 2, \dots\}$ is a martingale with respect to $\{X_1, X_2, \dots\}$ and T is a stopping time, then

$$E[Z_{T \wedge n}] = E[Z_1]$$

where $T \wedge n = \min(T, n)$.

Martingale Stopping Theorem

- Whenever we can justify the interchange of limit and expectation,

$$E[Z_1] = \lim_{n \rightarrow +\infty} E[Z_{T \wedge n}] = E \left[\lim_{n \rightarrow \infty} Z_{T \wedge n} \right] = E[Z_T].$$

Technical Conditions for the Interchange

- We return to the matter of justifying an interchange of limit and expectation as $n \rightarrow +\infty$.
- The most general conditions that guarantee this operation are that the random variable sequence $\{Z_{T \wedge n}, n = 1, 2, \dots\}$ be uniformly integrable in the sense that

$$\lim_{a \rightarrow +\infty} \sup_n E[|Z_{T \wedge n}| \cdot 1_{\{|Z_{T \wedge n}| \geq a\}}] = 0.$$

Technical Conditions for Interchange

The uniform integrability is typically difficult to verify. But we do have several simpler sufficient conditions to ensure it. Some of the most commonly used conditions are:

- **Condition 1:** $|Z_{T \wedge n}| \leq M$ for all n
- **Condition 2:** $E[Z_{T \wedge n}^2] \leq M$ for some constant M
- **Condition 3:**
 - (i) $E[T] < \infty$;
 - (ii) There exists a constant K such that $E[|Z_{n+1} - Z_n| | X_1, \dots, X_n] < K$, for $n < T$

The interchange conditions are very crucial for the martingale stopping theorem to hold!

Example: Gambler's Ruin Problem Revisited

- Consider an individual who starts at position k and at each step either moves 1 position to the right with probability 0.5 or 1 position to the left with probability 0.5. Assume that the successive movements are independent.
- Suppose that he will stop moving when he reaches either N or 0. What is the probability that he stops at N ?

Example: Expected Number of Steps

- Consider the preceding example. What is the expected number of steps this individual takes to reach N or 0 ?

Example: Unfair Games

- Now suppose that the individual moves to his right with a higher probability. That is, let

$$P(X_i = 1) = p, \quad P(X_i = -1) = q = 1 - p,$$

with $p > q$.

- Let

$$S_n = \sum_{i=1}^n X_i.$$

- Verify that $Z_n = (q/p)^{S_n}$ is a martingale. Using it and the martingale stopping theorem, we can compute the probability that he reaches N earlier than 0.

Financial Application

Consider a stock price S_n that is a martingale under the risk-neutral measure. Let τ be the first time S_n hits a certain barrier

Martingale Stopping Theorem allows us to compute the expected payoff:

$$\mathbb{E}[f(S_\tau)] = f(S_0) \quad (\text{under certain conditions})$$

Martingale Inequalities

Markov Inequality & Chebyshev Inequality

Markov Inequality. For a non-negative random variable X , for any $a > 0$

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Chebyshev Inequality: For any random variable X with mean μ and variance σ^2 ,

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2},$$

Weak Law of Large Number (WLLN)

Theorem

Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{E}[X_1] = \mu$ and $\text{Var}[X_1] = \sigma^2$. Define $S_n = \sum_{i=1}^n X_i$. Then for any $\varepsilon > 0$:

$$\lim_{n \rightarrow \infty} P \left(\left| \frac{S_n}{n} - \mu \right| \geq \varepsilon \right) = 0.$$

Equivalently:

$$\frac{S_n}{n} \xrightarrow{P} \mu \quad \text{as } n \rightarrow \infty.$$

Doob's Maximal Inequality

Theorem (Doob's Maximal Inequality)

Let $(M_n)_{n \geq 0}$ be a non-negative martingale. Then for any $a > 0$:

$$\mathbb{P} \left(\max_{0 \leq k \leq n} M_k \geq a \right) \leq \frac{\mathbb{E}[M_n]}{a}$$

Intuition:

- The probability that the maximum up to time n exceeds a is bounded by the expected value at time n divided by a .
- Similar to Markov's inequality, but for the maximum rather than just the final value.

Example: Stock Price Martingale

Suppose M_n is a non-negative martingale representing a stock price. Let $\mathbb{E}[M_n] = S$ be the expected price at time n . What is the probability that the stock price ever doubles S ?

Kolmogorov's Inequality

Theorem (Kolmogorov's Inequality)

Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}[X_i] = 0$ and $\text{Var}(X_i) < \infty$. Define $S_k = X_1 + \dots + X_k$. Then for any $\lambda > 0$:

$$\mathbb{P} \left(\max_{1 \leq k \leq n} |S_k| \geq \lambda \right) \leq \frac{\text{Var}(S_n)}{\lambda^2}$$

Comparison with Chebyshev:

- Chebyshev: $\mathbb{P}(|S_n| \geq \lambda) \leq \frac{\text{Var}(S_n)}{\lambda^2}$
- Kolmogorov: Same bound but for the **maximum** over the entire path
- Much stronger!

Example: Random Walk

Consider a simple symmetric random walk with $S_n = \sum_{i=1}^n \pm 1$, starting at 0. What is $\mathbb{P}(\max_{1 \leq k \leq 100} |S_k| \geq 20)$?

Strong Law of Large Numbers (SLLN)

Theorem

Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{E}[X_1] = \mu$ and $\text{Var}[X_1] = \sigma^2$. Define $S_n = \sum_{i=1}^n X_i$. Then:

$$P\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu\right) = 1.$$

Equivalently:

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu \quad \text{as } n \rightarrow \infty.$$

Borel–Cantelli

Let $A_1, A_2, \dots$ be events. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$$

then

$$\mathbb{P}(A_n \text{ occurs infinitely often}) = 0$$